

Odd Moore Spaces Give a Projective-Free Answer to a Hermiteness Question

Math discovery candidate full solution

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1 Status and source question

Brudnyi and Sasane ask whether there is a Hausdorff space X with $H^n(X; \mathbb{Z}) \neq 0$ for some $n \geq 5$ such that $C(X)$ is Hermite [1]. The question is printed on page 9 of the published-layout arXiv PDF:

In connection with Theorems 2.8, 2.9 the following question seems quite natural:

Question. *Is there a Hausdorff topological space X with $H^n(X, \mathbb{Z}) \neq 0$ for some $n \geq 5$ such that the algebra $C(X)$ is Hermite?*

The answer is yes. In fact, a five-dimensional finite CW complex already works, and its function algebra is projective free, which is stronger than Hermite. The same construction gives a sharp odd/even dichotomy.

2 Proof intuition

The source's sphere S^5 fails because the exceptional group $\pi_4(U(2)) \cong \mathbb{Z}/2$ produces a nontrivial stably trivial rank-two bundle. Replace the free top class of S^5 by odd torsion: attach a 5-cell to S^4 by an odd-degree map. The universal coefficient theorem moves the resulting $\mathbb{Z}/m$ homology into H^5 , so the required high-degree cohomology survives.

At the same time, the Puppe sequence says that vector bundles on this mapping cone are controlled by multiplication by m on just two homotopy groups of $BU(r)$. Odd multiplication is invertible on the troublesome $\mathbb{Z}/2$, while it is injective on the adjacent $\mathbb{Z}$. Exactness therefore leaves no nontrivial bundle in any rank.

3 The sharp Moore-space theorem

For an integer $m \geq 2$, let

$$X_m = M(\mathbb{Z}/m, 4) := S^4 \cup_{d_m} e^5, \quad (1)$$

where $d_m : S^4 \rightarrow S^4$ has degree m .

Theorem 1 (Parity dichotomy). *For every integer $m \geq 2$,*

$$H^5(X_m; \mathbb{Z}) \cong \mathbb{Z}/m.$$

Moreover:

1. if m is odd, every finite-rank complex vector bundle over X_m is trivial, and $C(X_m)$ is projective free;
2. if m is even, X_m supports a nontrivial stably trivial rank-two complex vector bundle, and $C(X_m)$ is not Hermite.

Proof. Cohomology. The reduced cellular chain complex of X_m is concentrated in degrees 5 and 4:

$$0 \longrightarrow \mathbb{Z} \xrightarrow{m} \mathbb{Z} \longrightarrow 0. \quad (2)$$

Thus $H_4(X_m; \mathbb{Z}) \cong \mathbb{Z}/m$ and $H_5(X_m; \mathbb{Z}) = 0$. The integral universal coefficient theorem gives

$$H^5(X_m; \mathbb{Z}) \cong \text{Ext}(H_4(X_m; \mathbb{Z}), \mathbb{Z}) \cong \mathbb{Z}/m. \quad (3)$$

Classifying all complex vector bundles. The cofibration defining (1) continues as

$$S^4 \xrightarrow{d_m} S^4 \longrightarrow X_m \longrightarrow S^5 \xrightarrow{\Sigma d_m} S^5. \quad (4)$$

For $Y = BU(r)$, its contravariant Puppe sequence contains the exact segment of pointed sets

$$\pi_5(Y) \xrightarrow{m} \pi_5(Y) \longrightarrow [X_m, Y] \longrightarrow \pi_4(Y) \xrightarrow{m} \pi_4(Y). \quad (5)$$

Here precomposition by d_m , or by its suspension, is multiplication by m on the corresponding homotopy group.

The required low homotopy groups are

r	$\pi_5(BU(r))$	$\pi_4(BU(r))$
1	0	0
2	$\mathbb{Z}/2$	$\mathbb{Z}$
$r \geq 3$	0	$\mathbb{Z}$.

(6)

Indeed, $\pi_k(BU(r)) = \pi_{k-1}(U(r))$; $\pi_4(U(2)) = \pi_4(S^3) = \mathbb{Z}/2$, while $\pi_4(U(r)) = 0$ for $r \geq 3$, and $\pi_3(U(r)) = \mathbb{Z}$ for $r \geq 2$ [3, 4]. Every $BU(r)$ is simply connected, so for the connected complex X_m the pointed homotopy sets in (5) agree with the unpointed homotopy sets that classify bundles.

Suppose first that m is odd. In every row of (6), the first map in (5) is surjective and the last map is injective. Exactness forces $[X_m, BU(r)]$ to consist only of its basepoint. Since rank- r complex vector bundles over the finite CW complex X_m are classified by $[X_m, BU(r)]$, every such bundle is trivial.

The space X_m is connected and compact Hausdorff. Proposition 2.1 of the source identifies triviality of all its finite-rank complex vector bundles with projective freeness of $C(X_m)$. This proves part (1).

Now let m be even and set $Y = BU(2)$. In (5), multiplication by m on $\mathbb{Z}/2$ is zero, whereas multiplication by m on $\mathbb{Z}$ is injective. Exactness shows that the middle pointed set has exactly two elements:

$$[X_m, BU(2)] \cong \mathbb{Z}/2. \quad (7)$$

The nonbasepoint class is a nontrivial rank-two bundle E .

After stabilization to BU , the corresponding Puppe sequence has $\pi_5(BU) = \pi_4(U) = 0$ and $\pi_4(BU) = \pi_3(U) = \mathbb{Z}$. Hence $[X_m, BU]$ is a singleton, so the stabilized class of E is null. Because X_m is a finite CW complex, this nullhomotopy occurs at a finite stage: for some k ,

$$E \oplus \varepsilon^k \cong \varepsilon^{k+2}. \quad (8)$$

Thus E is nontrivial but stably trivial. Proposition 2.3 of the source then says that $C(X_m)$ is not Hermite. $\square$

Corollary 2 (Full affirmative answer). *For*

$$X = M(\mathbb{Z}/3, 4) = S^4 \cup_{d_3} e^5,$$

one has $H^5(X; \mathbb{Z}) \cong \mathbb{Z}/3$, and $C(X)$ is projective free. In particular, $C(X)$ is Hermite.

Proof. Apply the odd case of Theorem 1 with $m = 3$. □

4 Checks, scope, and novelty

The pointed-set exactness in (5) is enough: injectivity of the last multiplication makes every class restrict trivially, and surjectivity of the first multiplication makes the preceding action have only the basepoint as its image. As an independent consistency check,

$$H^5(X_m; \mathbb{Z}/2) \cong \text{Ext}(\mathbb{Z}/m, \mathbb{Z}/2) = \begin{cases} 0, & m \text{ odd}, \\ \mathbb{Z}/2, & m \text{ even}, \end{cases}$$

which exactly matches the exceptional $\pi_4(U(2))$ term.

The result answers only the source’s existence question; it does not classify all Hausdorff spaces whose continuous-function algebras are Hermite. It does, however, classify the full Moore-space family (1) by parity and proves the stronger projective-free conclusion in the affirmative cases.

Bounded searches through August 17, 2026 used the exact question, exact title, arXiv id, and combinations of Moore spaces, complex vector bundles, Hermite algebras, and projective freeness. The published 2023 paper still states the question. Its two citing works indexed by OpenAlex concern bounded-variation and Dirichlet-series algebras, not this construction. No prior answer was located, so novelty is a candidate claim pending specialist review.

References

- [1] A. Brudnyi and A. Sasane, *Projective freeness and Hermiteness of complex function algebras*, Advances in Mathematics 434 (2023), 109322; arXiv:2208.04901.
- [2] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002.
- [3] A. Hatcher, *Vector Bundles and K-Theory*, Version 2.2, 2017.
- [4] M. Mimura and H. Toda, *Topology of Lie Groups, I and II*, Translations of Mathematical Monographs 91, American Mathematical Society, 1991.